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MAMMAL - Molecular Aligned Multi-Modal Architecture and Language for biomedical discovery



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Modern AI (Artificial Intelligence) methods offer new opportunities in pharmacology by enabling improved modeling of disease mechanisms and drug action learned from large and heterogeneous biological datasets. A central challenge is developing models that can jointly integrate disparate biomedical modalities. We introduce **MAMMAL** (**M**olecular **A**ligned **M**ulti **M**odal **A**rchitecture and **L**anguage), a foundation model for cross-modal learning, designed to address the challenges associated with drug discovery tasks. MAMMAL was pre-trained on 2 billion samples across protein and antibody sequences, small molecules, and gene expression profiles, and supports classification, regression, and generative tasks on cross-modal inputs. Across eleven benchmarks covering multiple stages of the drug discovery pipeline, MAMMAL achieves state-of-the-art performance on nine tasks and competitive results on two. In an antibody-antigen binding benchmark, fine-tuned MAMMAL prediction scores significantly outperform AlphaFold3 confidence scores, used here as a reference proxy for binding likelihood, in five of seven antigen targets. The MAMMAL framework and pretrained models are publicly available to support open and collaborative research.

Despite major advances in understanding health and disease, many complex biological processes remain poorly understood. Recent progress in artificial intelligence has raised the prospect that longstanding health challenges may prove treatable with AI-based methods¹ and that the high failure rate of drugs in clinical trials, where approximately 90% of drug candidates fail to achieve regulatory approval², can be improved. Over the past century, the translation of biological discoveries into effective therapies has matured. The drug discovery journey follows a multi-step pipeline that begins with identifying disease-associated proteins, progresses to finding compounds that can effectively target these proteins, and culminates in optimizing drug candidates to meet stringent standards for efficacy and safety. This process is both costly and labor-intensive, requiring extensive laboratory assays that measure drug-target interactions, characterize cellular responses, and validate therapeutic safety and efficacy^{3–5}. Therapeutic modalities range from small molecules, which are stable, easy to manufacture, and suitable for oral delivery⁶, to biologic therapeutics, such as engineered antibodies, which offer high specificity but require more complex manufacturing and delivery⁷.

Accelerating drug discovery and improving the predictive accuracy of its steps have become a central focus in biomedical research, with the goal of streamlining target identification, drug design, and testing^{8–10}. Analyzing gene expression profiles, particularly from single-cell RNA sequencing (scRNA-seq), has emerged as a key approach for distinguishing disease-associated cell populations and characterizing drug-induced cellular responses^{11–14}. These analyses support target identification, mechanistic understanding, and assessment of drug effects across diverse cellular contexts^{15,16}. In parallel, generative models have been increasingly applied to drug design, enabling the synthesis of novel candidate molecules for further evaluation^{17–19}. As high-throughput screening assays for measuring drug binding affinity are costly and challenging to scale, accurate prediction of drug-target interactions can significantly enhance drug design, improving both efficacy and precision. Overall, predictive modeling of binding affinity, toxicity, and efficacy in the early stages of the pipeline can reduce reliance on expensive late-stage testing, ultimately saving time and resources in drug development.

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A central challenge in developing predictive and generative AI models for drug discovery is determining how the different biomedical modalities and entities should be represented and combined as model inputs and outputs²⁰. This challenge is particularly acute for interaction-centric tasks, such as predicting whether a small-molecule drug will bind a specific protein target. While sequence-based representations of molecules (e.g., SMILES) and proteins have demonstrated considerable success, the representation of transcriptomic data as sequences has only recently been explored, leading to a range of emerging approaches²¹.

In this work, we introduce a cross-domain foundation model that represents small molecules, proteins, and transcriptomic data within a unified sequence framework. We present MAMMAL (Molecular Aligned Multi-Modal Architecture and Language), designed to address core challenges in AI-driven drug discovery. Our key contributions are: (1) a novel multi-alignment framework that integrates molecular, protein, and gene-expression inputs using a flexible, structured prompt syntax, enabling modeling of complex multi-entity interactions such as drug-target binding; (2) an architecture tailored to drug discovery that natively incorporates numerical values into its embedding space via continuous projections. This improves precision for tasks such as affinity prediction and structural integration, while supporting both predictive and generative objectives; and (3) a large-scale pre-trained model trained on over 2 billion samples across denoising, infilling, and binding-related tasks. Across 11 benchmarks spanning multiple stages of the drug discovery pipeline, fine-tuned MAMMAL achieves state-of-the-art performance on nine tasks. The model and pre-trained weights are publicly available on <https://huggingface.co/ibm/biomed.omics.bl.sm.ma-ted-458m> under the name `ibm/biomed.omics.bl.sm.ma-ted-458m`, providing a platform for advancing research in multimodal biomedical modeling.

Results

To evaluate the performance and generalization capabilities of `ibm/biomed.omics.bl.sm.ma-ted-458m`, we selected a diverse set of existing benchmarks spanning multiple task types and stages of the drug discovery pipeline, prioritizing benchmarks with clearly defined splits when those were available. We assessed model quality through a fine-tuning-based evaluation strategy, where the pretrained model is adapted to each benchmark and compared against specialized state-of-the-art (SOTA) models. The evaluation methodology and fine-tuning protocol as well as detailed descriptions of each benchmark—including background, significance for drug discovery, prior models, and data statistics—are provided in the subsections below. A summary of performance across tasks is presented in Table 1 and visualized in Fig. 1E, and

representative encoder-decoder examples are provided in the Supplementary Table S1.

AlphaFold²², whose development contributed to the 2024 Nobel Prize in Chemistry, revolutionized protein structure prediction. Its extension AlphaFold-Multimer²³ enabled modeling of antibody-antigen complexes, while AlphaFold 3 (AF3)²⁴ further improved accuracy and added nucleic acid/small molecule support. Motivated by AF3's reported advances, we evaluated its performance on therapeutic Antibody and Nanobody complexes (Subsection 2.10). Comparative analysis reveals that MAMMAL achieves better classification performance than AF3 in five of seven targets (Table 2).

Evaluation

We compiled a comprehensive set of 11 benchmarks covering multiple data domains and task types, including classification, regression and generation, as well as single-entity, multi-entity, and multi-domain tasks. These benchmarks address key stages of the drug discovery process: Identifying target cell types (Cell Type) and advancing precision medicine (Cancer-Drug Response 1-3); predicting drug efficacy (BBBP) and safety (ClinTox); predicting the binding affinity of small-molecule drugs to target proteins (DTI); predicting interactions of biological drugs (PPI); and designing new drugs, such as antibodies, to target specific proteins (Ab Infilling).

To enable fair and direct comparison to prior work, benchmark selection prioritized datasets with predefined train, validation, and test splits or with established splitting strategies reported in the corresponding state of the art studies. For each benchmark, we followed the data splits and evaluation metrics used in the original benchmark or SOTA reference. When explicit train-validation-test splits were available, `ibm/biomed.omics.bl.sm.ma-ted-458m` was fine tuned on the training set, the best checkpoint was selected using the validation set, and final performance was reported on the test set. For benchmarks evaluated using cross validation, we adopted the same protocol as the corresponding prior work. Unless otherwise noted, standard errors were estimated by training the models with three different random seeds and calculating the standard deviation of their performance on the held out test set. Detailed descriptions of each benchmark, the fine tuning procedures, and the evaluation protocols are provided alongside the corresponding results. For the DTI benchmark, performance is reported using normalized root mean square error (NRMSE), defined as the root mean square error divided by the standard deviation of the test labels. The same normalization is applied to both MAMMAL and reported SOTA results, yielding values below 1 and enabling joint visualization alongside other performance metrics in Fig. 1(E). We consider MAMMAL to outperform existing state of the art when the relative improvement, computed as $[\text{SOTA} - \text{MAMMAL}]/\text{SOTA}$, exceeds 1%.

Table 1 | Comparison of SOTA and MAMMAL Performance Across Benchmarks

Benchmark	Domain	Type	Metric	SOTA	MAMMAL	Imp.
Cell type	GE	cls	↑ F1	27 0.710	0.763 ± 0.012	7.5 %
BBBP	SM	cls	↑ AUROC	32 0.937	0.957 ± 0.006	2.2 %
ClinTox	SM	cls	↑ AUROC	32 0.948	0.986 ± 0.007	4.0 %
Cancer-Drug Response 1	GE+SM	reg	↑ Pearson	92 0.887	0.917 ± 0.001	3.4 %
Cancer-Drug Response 2	GE+SM	reg	↑ Pearson	92 0.900	0.931 ± 0.002	3.4 %
Cancer-Drug Response 3	GE+SM	reg	↑ Pearson	36 0.923 [0.917–0.929]	0.928 ± 0.000	0.5 %
Ab Infilling	Protein	gen	↑ CDRH3-AAR	47 0.375	0.446 ± 0.002	19.0 %
AbAg Bind	Protein	cls	↑ AUROC	63 0.924 [0.923–0.925]	0.928 ± 0.002	0.4 %
TCR Bind	Protein	cls	↑ AUROC	54 0.862 [0.85–0.868]	0.879 ± 0.003	2.0 %
PPI ΔΔG	Protein	reg	↑ Pearson	104 0.663	0.852 ± 0.041	28.5 %
DTI	Prot.+SM	reg	↓ NRMSE	60 0.942 ± 0.028	0.906 ± 0.011	3.8%

For NRMSE, lower is better. For other metrics (AUROC, CDRH3-AAR, Pearson, Spearman, and F1), higher is better. Each row shows the results from a MAMMAL model fine-tuned from `ibm/biomed.omics.bl.sm.ma-ted-458m` for the corresponding task. In the "Type" column: *cls* classification, *reg* regression, *gen* generation, *Imp.* improvement (percentage) of our model over SOTA. In the "Domain" column: *GE* genes expression, *SM* small molecule, *Prot.* protein.

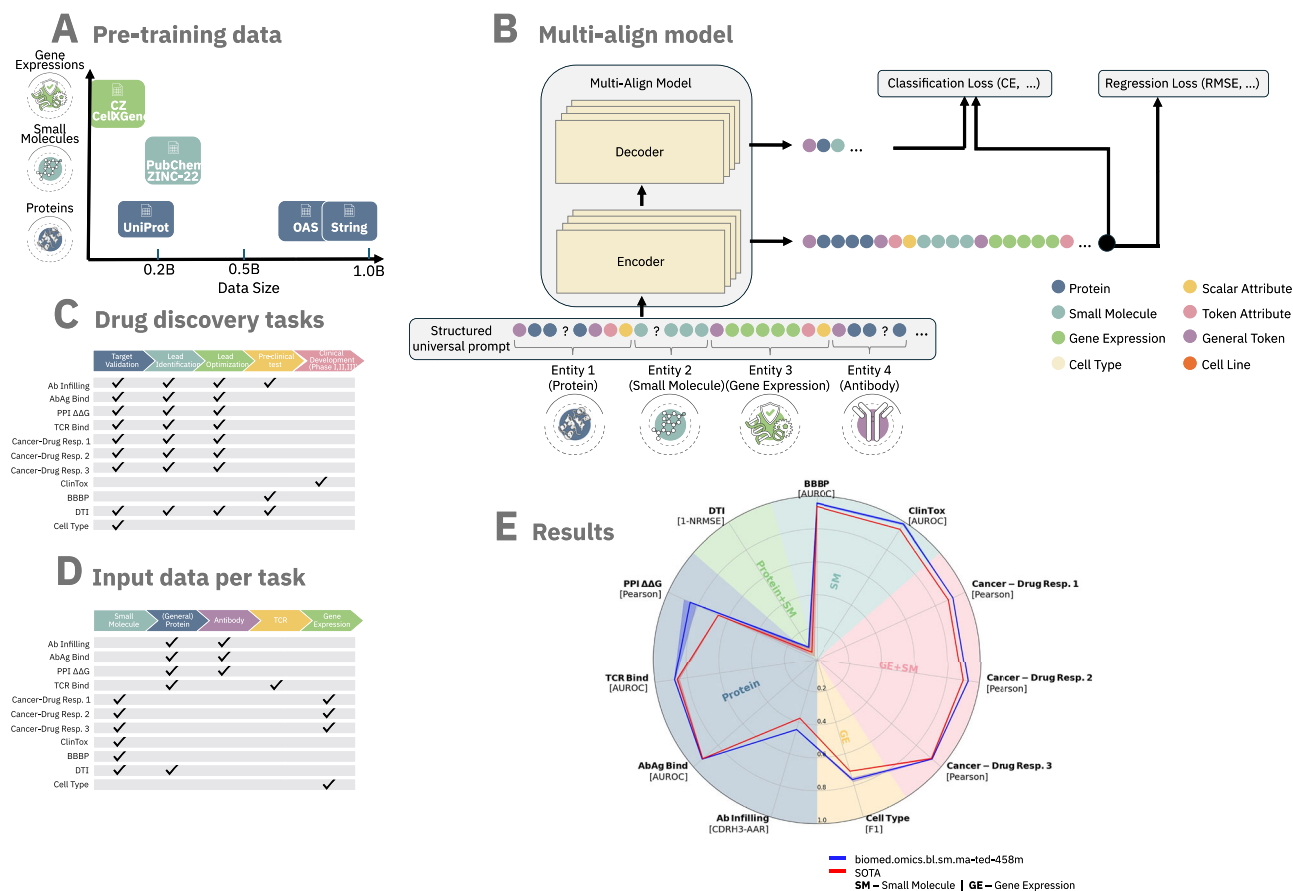


Fig. 1 | Overview of MAMMAL pretraining data, model architecture, and downstream tasks. **A** We introduce a multi-align model pretrained on six datasets, each containing tens to hundreds of millions of data points. These data points include protein sequences, small molecules, and gene expression profiles, with a combined sample size of 2 billion. **B** The multi-align model combines flexible encoder-only and encoder-decoder components. It takes sequences as input, which may contain any combination of tokens and scalar elements, processed by an encoder stack consisting of self-attention blocks. In encoder-only mode, a dedicated token prediction head outputs logits for token predictions, with an optional scalar

prediction head for scalar outputs. In encoder-decoder mode, residual connections inject features from the encoder's final hidden layer into each decoder layer, and a decoder-specific prediction head outputs the final logits. **C** Diverse downstream tasks performed by the multi-align model, mapped to their contributions within the steps of a typical drug discovery pipeline. **D** Diverse downstream tasks performed by the multi-align model, categorized by data type used in the fine-tuning process. **E** Performance of the multi-align model across a diverse set of tasks compared to SOTA. Panel (E) was generated using Matplotlib. Panels (A–D) were created using Illustrator and PowerPoint.

Table 2 | Per-target AUROC comparison between MAMMAL and AF3 on held-out antibody/nanobody–antigen test subsets

Target	length (AA)	<i>n</i>	pos	MAMMAL AUROC	AF3 AUROC	ΔAUROC	<i>P</i> -value
HER2 ECD	630	60	30	0.93	0.45	+0.42	1.5×10^{-6}
Albumin	609	28	11	0.91	0.59	+0.32	7.4×10^{-3}
CD206	1,456	37	14	1.00	0.59	+0.41	1.1×10^{-5}
EGFR	1,210	28	11	0.94	0.49	+0.45	2.2×10^{-5}
TBG	384	32	5	0.63	1.00	−0.37	1.0×10^{-4}
TNFα	233	34	13	0.86	0.87	−0.01	NS
VWF	2,813	34	13	0.83	0.32	+0.51	3.9×10^{-6}

P-values were computed using the DeLong test for correlated ROC curves. ΔAUROC denotes $\text{AUROC}_{\text{MAMMAL}} - \text{AUROC}_{\text{AF3}}$. NS indicates a non-significant difference ($p > 0.1$). ECD denotes the extracellular domain. Bold in the AUROC column of a method (MAMMAL or AF3) indicates higher result for that method. Bold in the ΔAUROC column indicates rows where MAMMAL outperforms AF3.

Cell Type Annotation

Cell type prediction enables researchers to distinguish between different cell populations, such as those associated with various diseases^{11–14}. It is also crucial for understanding how diseases or drugs affect different cell types. In recent years, a variety of methods have been developed for this task, including approaches based on marker genes, correlation-based techniques,

and annotation using classification²⁵. Recent advances in transformer-based and large-scale foundation models^{26–28} have shown improved performance.

The input for this task is single-cell gene expression data. The benchmark we used was based on the Zheng68k dataset²⁹, which is composed of human peripheral blood mononuclear cells and is widely used for evaluating cell-type annotation performance, due to the

similarity of the cell types involved. The dataset contains 68,579 cells across 11 cell types and originally included 32,738 genes, which after removing non-expressed genes leaves 20,387 genes in the benchmark. Preprocessing involved normalization, log transformation of expression values, followed by binning. Similar to the approach in³⁰, our model uses a ranked list of expressed gene names, ordered by their expression levels, as input. The label to predict is provided in the cell ontology format “CL:NNNNNN” (see Supplementary Table S1).

Following prior work²⁷, we adopted a 5-fold cross-validation strategy to fine-tune and evaluate `ibm/biomed.omics.bl.sm.ma-ted-458m`, ensuring similar proportions of cell types across folds, and assessed performance using accuracy and macro F1 score. MAMMAL outperforms the previous state-of-the-art performance in both accuracy and F1 (Table 1 and detailed results in Supplementary Table S2), achieving a 7.5% improvement in F1.

BBBP and ClinTox

To ensure the development of safe and effective drugs, candidates must satisfy rigorous criteria related to both efficacy and safety. In this study, we selected two relevant benchmarks from MoleculeNet³¹, a widely used suite of benchmarks for evaluating machine learning models on small-molecule drug properties: BBBP and ClinTox. The BBBP benchmark focuses on predicting the ability of drugs to penetrate the blood-brain barrier, a critical consideration for drugs targeting the central nervous system. The ClinTox benchmark comprises two related tasks: (1) predicting failure in clinical toxicity trials, and (2) predicting FDA approval status. The overall performance on ClinTox is reported as the average performance across these two tasks.

MolFormer³², a well-established model for molecular embeddings trained on 1.1 billion SMILES sequences, has achieved state-of-the-art performance on both the BBBP and ClinTox benchmarks. In our study, we adopted the benchmarks from³², which provided predefined splits for training, validation, and testing. MAMMAL surpasses MolFormer on both benchmarks (Table 1), achieving an average area under the receiver operating characteristic curve (AUROC) score of 0.957 on BBBP and 0.986 on ClinTox, representing improvements of 2.2% and 4%, respectively, over the state of the art.

Cancer-Drug Response

Identifying drug response at the cellular level is a critical step in the development of new drugs. Two key public databases supporting this effort, particularly in cancer drug development, are the Cancer Cell Line Encyclopedia (CCLE)³³ and the Genomics of Drug Sensitivity in Cancer (GDSC)³⁴. CCLE provides multi-omics profiles for around 1000 cancer cell lines, while GDSC offers data on the drug responses of these lines to hundreds of drugs, commonly measured using the half-maximal inhibitory concentration (IC₅₀). Notable computational models addressed this task^{35–37}.

For our study, we used three subsets of the GDSC database: GDSC1 and GDSC2, available through the Therapeutics Data Commons (TDC)³⁸, and referred to in the paper as Cancer-Drug Response 1 and Cancer-Drug Response 2, respectively; and a subset published in³⁶, referred to as Cancer-Drug Response 3. A dataset statistics table summarizing the number of cell lines, drugs, and cell–drug pairs is provided in Supplementary Table S3. We used the random splits provided by TDC for Cancer-Drug Response 1 and 2, while for Cancer-Drug Response 3, we followed the split methodology outlined in³⁶, reserving 5% of the data for the test set, stratified by TCGA³⁹ pathways associated with the cancer cell lines.

During fine-tuning, we used only gene-expression profiles and SMILES representations of drugs, as shown in the example prompt in the Supplementary Table S1. Similar to the input format for cell type annotation, gene-expression profiles were provided as ranked lists of gene names based on their expression levels. For predicting continuous IC₅₀ values, MAMMAL was utilized in regression mode, taking advantage of its built-in support for floating-point scalar predictions. Our model outperforms the

current SOTA models for Cancer-Drug Response 1 and 2 (Table 1), achieving a 3.4% increase in Pearson correlation values. Additionally, it yields results comparable to the SOTA for the Cancer-Drug Response 3 benchmark, with a slight improvement of 0.5%.

To further evaluate MAMMAL’s predictive capability on novel compounds, we assessed drug response predictions for four drugs not present in the GDSC training data: Carfilzomib, Nintedanib, Infigratinib, and Vemurafenib. Tanimoto similarity analysis confirmed that three of these drugs (Carfilzomib, Nintedanib, and Infigratinib) have no structurally similar compounds in the training set (Tanimoto coefficient <0.7), while Vemurafenib shares moderate similarity (0.82) with PLX-4720, a BRAF inhibitor present in GDSC. We performed experimental validation using the same assay protocol employed in GDSC: cell viability was measured using CellTiter-Glo following 72-hour drug incubation, and IC₅₀ values were determined with Prism (GraphPad). The experimental measurements revealed a consistent potency ranking across all tested cell lines: Carfilzomib (most potent), followed by Nintedanib, Infigratinib, and Vemurafenib (least potent). MAMMAL predictions reproduced this exact ranking for the tested cell lines. When extended to all 805 cell lines in GDSC, the model preserved this relative ordering in approximately 90–95% of cases, suggesting that the predicted potency differences are largely cell line-independent.

Notably, Carfilzomib is a proteasome inhibitor approved exclusively for hematological malignancies (multiple myeloma), with limited efficacy in cells of solid tumors⁴⁰. The model’s prediction of Carfilzomib as the most potent agent across diverse solid tumor cell lines aligns with our experimental observations and suggests potential broader applicability that warrants further investigation.

Ab Infilling

Antibodies are a family of proteins produced by the immune system to neutralize foreign antigens and are of particular interest due to their high specificity and strong binding to target molecules^{41,42}. These characteristics have made them a crucial class of therapeutics, driving significant research efforts into the design of new antibody-based drug candidates^{7,43–45}. Antigen-binding fragments (Fabs) are the antibody fragments that bind to antigens. It is composed of one constant domain and one variable domain of each of the *heavy* and *light* chains. Each variable region is further divided into four framework (FR) regions and three complementarity-determining regions (CDRs). While FR regions are typically conserved, CDRs exhibit significant variation in their amino acid composition and are generally the primary determinants of binding affinity to the target antigen. When designing novel antibodies for a specific antigen, the typical approach is to explore alternative CDRs that could produce a new, functional antibody with high binding affinity to the target^{41,42,46,47}.

Recently, several deep learning methods have been developed for targeted antibody design, framing CDR prediction as an *infilling* task^{46–52}. These models predict missing CDR regions, represented by MASK tokens, using the amino acid sequences of both the antigen and the antibody’s FR regions. While prior approaches often relied on structural data, this information is scarce and challenging to obtain⁵³. In contrast, we fine-tune MAMMAL for the targeted antibody design task using only the sequence data of the antigen and the sequence of the antibody’s FR regions.

The targeted antibody design task benchmark is based on the SAbDab dataset⁵³. Following the data processing outlined in⁴⁷, we filtered out samples with missing CDRs to enable direct comparison, even though MAMMAL supports samples that contain missing CDRs. Consistent with⁴⁷, we randomly partitioned the dataset into training, validation, and test folds while ensuring that samples with similar heavy-chain third CDR (CDRH3) subsequences remained in the same fold. MAMMAL demonstrates superior amino acid recovery (AAR), defined as the fraction of correctly predicted residues, across all masked CDRs (Table 1; detailed results are provided in Supplementary Table S4). Notably, in CDRH3, the most variable region, it exhibits a remarkable improvement of 19%.

T-Cell Receptor-Epitope Binding

T-cell receptor (TCR) binding to immunogenic peptides (epitopes) presented by major histocompatibility complex molecules is a critical mechanism in the adaptive immune system, essential for antigen recognition and triggering immune responses. The TCR repertoire exhibits considerable diversity, consisting of an α -chain and a β -chain that function together to enable T cells to recognize a wide array of epitopes. The β -chain is especially significant, as it is crucial for the early stages of T-cell development and possesses greater variability, which enhances the TCR's capacity to identify diverse pathogens effectively. However, understanding the specific interactions between TCRs and epitopes remains a significant challenge due to the vast variability in TCR sequences. Accurate prediction of TCR-peptide binding from sequence data would advance immunology by offering deeper insights into a patient's immune status and disease history. This capability holds potential applications in personalized immunotherapy, early diagnosis, and the treatment of diseases such as cancer and autoimmune disorders. In silico tools designed to model TCR-peptide interactions could facilitate the study of therapeutic T-cell efficacy and assess cross-reactivity risks, presenting an opportunity for precision medicine.

We evaluated the model on the task of predicting TCR-epitope binding from sequence data using the Weber benchmark (ref. 54, https://tdcommons.ai/multi_pred_tasks/tcrepitope), which consists of 47,182 TCR β -chain epitope pairs. This dataset covers 192 distinct epitopes and includes 23,139 unique TCR β -chain sequences, with 50% of the pairs serving as negative samples created by randomly pairing TCR sequences with epitopes they are not known to bind with. The dataset also includes the CDR3 subsequence for each TCR β -chain, the most hypervariable region of the chain. We used 10-fold cross-validation. The folds were pre-defined in⁵⁴. Fine-tuning involved three concurrent tasks: TCR β -chain mask infilling and two classification tasks: (i) TCR β -chain epitope binding prediction and (ii) TCR β -chain -CDR3 epitope binding prediction. Here, we report the performance only for the TCR β -chain epitope binding prediction task. Our model achieves an average AUROC of 0.879 (Table 1), representing a statistically significant improvement of 2% over the SOTA, as our result falls outside the SOTA's confidence interval.

Protein-Protein Interaction - $\Delta\Delta G$ Prediction

An important factor in drug design is binding affinity, commonly measured by the equilibrium dissociation constant, K_D , which is related to the Gibbs free energy ΔG through the equation

$$\Delta G = kT \ln(K_D), \quad (1)$$

where k is the Boltzmann constant and T is the temperature⁵⁵.

The effect of introducing mutations into a protein-protein complex is commonly quantified by the change in binding free energy relative to the reference (wild-type) complex. This mutation-induced effect is captured by the difference in Gibbs free energy, defined as

$$\Delta\Delta G = \Delta G_{\text{mutant}} - \Delta G_{\text{wild-type}}.$$

By subtracting the wild-type free energy, $\Delta\Delta G$ isolates the energetic contribution of the mutation itself. As a result, $\Delta\Delta G$ provides a direct measure of whether a mutation stabilizes or destabilizes binding and is a standard target in studies of mutational effects on protein-protein interactions^{56–58}.

The SKEMPI dataset⁵⁵ provides experimentally measured changes in thermodynamic parameters, including ΔG and kinetic rate constants, for mutations in protein-protein complexes with known structures in the Protein Data Bank⁵⁹. This dataset is widely used to benchmark methods for predicting mutation-induced changes in binding affinity, particularly $\Delta\Delta G$. A commonly used subset of SKEMPI comprising 1131 single-point mutations (S1131) is adopted as our benchmark. Following standard practice, we report 10-fold cross-validation performance on this subset. The input for our model consists solely of amino acid sequences for the wild-type and

mutant complexes, without structural information. Leveraging MAMMAL-s support for continuous-valued outputs, we formulate $\Delta\Delta G$ prediction as a regression task. Performance results are reported in Table 1. Our model achieves an average Pearson correlation of 0.852, substantially exceeding the previous sequence-only state of the art (0.663), and remains competitive with structure-based methods, falling only 1.6% short of the reported best performance of 0.866⁵⁶.

Drug-Target Interaction

Predicting drug-target binding affinity plays a crucial role in the early stages of drug discovery. Traditionally, binding affinities are measured through high-throughput screening experiments, which, while accurate, are resource-intensive and limited in their scalability to evaluate large sets of drug candidates. In this task, we focus on predicting binding affinities using pK_D , the negative logarithm of the dissociation constant, which reflects the strength of the interaction between a small molecule (drug) and a protein (target). We utilize the PEER (Protein sEquence uNDERstanding) benchmark⁶⁰ for DTI prediction. This benchmark leverages data from the BindingDB dataset⁶¹, with a specific test split that holds out four protein classes - estrogen receptor, G-protein-coupled receptors, ion channels, and receptor tyrosine kinases - for assessing generalization performance on unseen classes.

For model fine-tuning, we conducted hyperparameter optimization, selecting an initial learning rate of 0.0004, with no dropout and no weight decay. We standardized the pK_D values based on the mean and standard deviation of the training set. For evaluation, we transformed the predicted values back to their original scale. Our model achieves an average NRMSE of 0.906 (Table 1), demonstrating a solid improvement of 3.8% over the SOTA reported by⁶⁰.

Antibody-Antigen Binding Prediction

Accurate prediction of antigen-antibody binding can enhance the design and optimization of therapeutic antibodies, leading to improved efficacy and specificity. We employ the human epidermal growth factor receptor 2 (HER2) dataset⁶² as a benchmark for predicting antibody-antigen binding. HER2 is a key target for certain types of breast and stomach cancers. The dataset includes variations of the clinically approved therapeutic antibody trastuzumab and their corresponding affinities for the HER2 antigen. The dataset comprises 8,935 binding and 25,114 non-binding trastuzumab CDR H3 mutants, each with up to 10 mutations, following de-duplication and the removal of samples labeled as both binding and non-binding.

For the most accurate comparison with the SOTA (refs. 62,63), the HER2 dataset was divided into train (70%), validation (15%) and test (15%) sets. For increased robustness, the train set was further divided into 5 folds. The reported results are from the 5 models trained on different train folds, and evaluated on the test set.

Finetuning involved feeding the target antigen sequence as well as the entire heavy-chain variable region as input and predicting binding to the target sequence. Our model achieves an average AUROC of 0.928 (Table 1), slightly surpassing the SOTA, which incorporated structural data, unlike our model.

Comparison of AlphaFold 3 and MAMMAL in Predicting Antibody-Antigen and Nanobody-Antigen Binding

Accurate prediction of antibody-antigen and nanobody-antigen interactions is essential for evaluating therapeutic efficacy and guiding protein engineering. Although AlphaFold 3 (AF3) is not explicitly designed as a binary protein-protein interaction (PPI) classifier, binding likelihood can be inferred from structure-derived confidence scores, such as predicted template modeling (pTM) and interface predicted template modeling (ipTM), computed from predicted protein-protein complexes. These confidence scores are derived from structural predictions rather than classification objectives. Recent studies suggest that these scores correlate with true binding events^{64,65}.

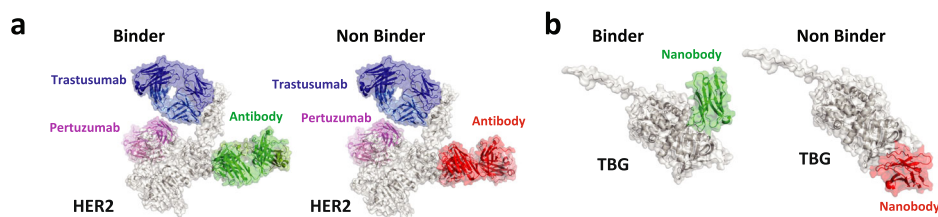


Fig. 2 | AF3-predicted nanobody binding poses on HER2 and TBG. a HER2 extracellular domain (ECD) structure with representative AF3-predicted complexes for a binder and a non-binder. The FDA-approved therapeutic antibodies trastuzumab (blue) and pertuzumab (purple) are shown for reference. AF3 predicts both binders and non-binders engaging the same region of the HER2 ECD, which is distinct from the known therapeutic epitopes, consistent with its poor discriminative

performance on this target (AUROC = 0.45). **b** Thyroxine-binding globulin (TBG) structure with AF3-predicted complexes for binding and non-binding VHHs. In contrast to HER2, AF3 predicts distinct binding poses for binders versus non-binders on TBG, consistent with its strong discriminative performance on this target. Visualizations were generated using PyMOL.

Accordingly, we conduct an exploratory comparison between AF3-derived confidence scores and a fine-tuned MAMMAL model for distinguishing binders from non-binders. We emphasize that AF3 provides detailed 3D structural hypotheses, whereas MAMMAL is a sequence-only model that produces probabilistic binding predictions; the comparison is intended to assess relative discriminative power for binding prediction rather than to equate the underlying modeling approaches.

We first evaluated the extracellular domain (ECD) of HER2, a well-characterized therapeutic antigen with experimentally validated binding epitopes. We used the HER2-specific MAMMAL model described in Subsection 2.9. Due to the computational demands and limited availability of AF3, the HER2 benchmark test set was downsampled to 60 examples, comprising 30 binders and 30 non-binders. The HER2-specific MAMMAL model demonstrates strong discriminative performance, achieving an AUROC of 0.88. In contrast, AF3 exhibits no meaningful separation between binders and non-binders (AUROC = 0.45), and the difference in performance between the two models is highly significant (DeLong test, $P = 1.5 \times 10^{-6}$). Structural analysis further reveals that AF3-predicted binding sites are indistinguishable between binders and non-binders and deviate from the known epitopes of the FDA-approved antibodies trastuzumab and pertuzumab (Fig. 2a). An extended comparative analysis of MAMMAL (pre-trained and fine-tuned) and AF3 is provided in Supplementary Table S5. This includes several AF3 confidence-score variants, using ipTM- and pTM-based scoring, and heavy-chain-only and heavy+light-chain input configurations.

Next, we evaluated nanobody binding across six structurally diverse antigen targets: albumin, mannose receptor (CD206), epidermal growth factor receptor (EGFR), thyroxine-binding globulin (TBG), tumor necrosis factor alpha (TNF α), and von Willebrand factor (VWF). Binding nanobodies were collected from SabDab-nano⁶⁶, patents, and proprietary datasets. Non-binders consisted of nanobodies experimentally confirmed as non-binding in phage-display library screenings, as well as nanobodies targeting unrelated antigens. From a total of 668 nanobody-antigen pairs (131 binders and 537 non-binders), we selected 475 sequences (64 binders and 411 non-binders) for MAMMAL fine-tuning and reserved 193 sequences (67 binders and 126 non-binders) for held-out evaluation. A single MAMMAL model was fine-tuned on training data comprising binders and non-binders across all targets, and subsequently evaluated separately on the test subset corresponding to each target, with performance compared against AF3 confidence scores computed on the same samples. Test set statistics and per-target MAMMAL and AF3 performance are summarized in Table 2. Additional performance metrics for MAMMAL and AF3 are presented in Supplementary Tables S6–S7. As shown, MAMMAL significantly outperforms AF3 on the larger targets: albumin, CD206, EGFR, and VWF. In contrast, AF3 achieves superior performance on the smaller TBG target, and analysis of the predicted structures highlights distinct binding sites for binders versus non-binders (Fig. 2b and Supplementary Figure S4). For the smallest protein, TNF α , the two models exhibit comparable performance.

Discussion

Recent progress in biomedical foundation models has enabled substantial yet largely domain-specific advances across cellular and molecular biology. Specialized model classes including RNA expression-based models^{26,28,30}, protein language models^{57,68}, and frameworks for small-molecule-protein interaction modeling³² provide increasingly detailed representations of biological systems. Together, these efforts suggest the feasibility of integrative frameworks such as the proposed AI-enabled “virtual cell”^{69,70}. Realizing this vision, however, requires approaches that can coherently align heterogeneous biological modalities within a scalable and unified representation.

MAMMAL advances this goal by reframing diverse biological tasks as sequence-to-sequence problems and introducing design choices tailored to drug discovery applications. Its multi-domain syntax and extensible tokenizer allow heterogeneous biological inputs and outputs to be expressed within a shared sequence representation. The implementation of continuous embeddings for numerical variables preserves quantitative precision without the information loss typical of discretization, facilitating integration of biochemical and pharmacological measurements. MAMMAL further supports both encoder-only and encoder-decoder configurations, offering the architectural flexibility required to address the broad spectrum of predictive and generative tasks encountered throughout the drug discovery pipeline. The strong performance of fine-tuned models across multiple domains and stages highlights the practical utility of this unified framework in established *in silico* benchmarks.

In addition to benchmark evaluations, MAMMAL was assessed in real-world settings, where it accurately predicted the relative potency ranking of four drugs, including Carfilzomib⁴⁰, a proteasome inhibitor approved only for hematological malignancies, across solid tumor cell lines. Wet-lab validation using the same CellTiter-Glo assay protocol as GDSC confirmed the predicted ordering. This result demonstrates the model’s capacity to generalize to novel compounds and identify potentially repurposable agents for indications beyond their current approval. In parallel, MAMMAL has been employed in joint studies with academic and industry collaborators, including a published investigation supporting its ability to predict antibody activity against influenza hemagglutinin proteins⁷¹. Additional collaborative studies involving prospective wet-lab validation of MAMMAL-guided predictions are currently in progress, further evaluating the framework across diverse biological and therapeutic contexts.

We further evaluated MAMMAL for antibody binder versus non-binder identification and compared it to AlphaFold 3 (AF3), which has recently been explored as a proxy for estimating binding propensity via predicted interfaces^{64,65}. Fine-tuned MAMMAL achieved robust discriminative performance across six of seven antigens including flexible, glycosylated, and multi-domain targets such as CD206, EGFR, HER2, albumin, and VWF with AUROCs ranging from 0.83 to 1.00. In contrast, AF3, applied in a zero-shot setting, showed limited ability to separate binders from non-binders for most targets (AUROC $\leq$ 0.59), performing well only on the rigid globular antigen TBG and matching MAMMAL on TNF α .

These limitations may reflect AF3's bias toward true-positive PDB complexes, its lack of explicit negative supervision, and its reliance on a single static conformation. As a result, AF3 struggles with proteins containing heterogeneous or intrinsically disordered regions (IDRs), which comprise 30–40% of the human proteome and are known to challenge structure-based models⁷², including IDR-rich therapeutic targets such as EGFR⁷³ and HER2⁷⁴. In contrast, protein language models (PLMs) such as MAMMAL better capture the statistical properties of IDRs, consistent with observations that pLM-based approaches perform well on IDR-related benchmarks⁷⁵. These factors together likely contribute to MAMMAL's improved robustness for binder identification across therapeutically relevant targets.

Text-based large language models excel at understanding and reasoning over human language and have been adapted to the biomedical domain through pretraining on PubMed abstracts and full-text articles. However, these models operate primarily on unstructured text and do not explicitly represent structured biological modalities. In a comparison with BioMedLM⁶, reported in the Supplementary Information, MAMMAL outperformed text-only models on tasks involving proteins and gene expression data, highlighting the limitations of language-only pretraining for biological data lying outside natural language, such as protein sequences and gene expression measurements. Beyond text-based LLMs, prior work includes graph neural networks, which encode molecular, protein, and biological-system topology and can directly represent knowledge graphs⁷⁷. While sequence-based representations can implicitly capture aspects of three-dimensional structure⁷⁸, explicitly modeling 3D information can further improve performance, motivating sequence models that input or generate structural representations^{79–81}. More recently, diffusion models have emerged as powerful methods for modeling molecules and proteins directly in continuous atomic coordinate space^{24,82–84}. In contrast to these architecture- or modality-specific approaches, MAMMAL adopts a unifying sequence-to-sequence abstraction that decouples task definition from architectural design. A key open challenge is how to combine this simplicity and flexibility with the expressive power of specialized models, particularly diffusion-based architectures that excel at high-fidelity structural generation.

The growing adoption of agentic workflows-leveraging LLMs and specialized agents to automate and accelerate discovery processes-marks a broader shift in biomedical research practice^{85,86}. Fine-tuned MAMMAL models integrate naturally into this paradigm, providing modular, task-specific components that support scalable and efficient discovery workflows. Fair and meaningful comparisons across biomedical models require benchmarks with accessible datasets, reproducible data splits, and clear evaluation protocols. Although the benchmarks used here satisfy these criteria and span multiple stages of the drug discovery pipeline, our evaluation is limited to models that publicly report results on these tasks, and claims of state-of-the-art performance should therefore be interpreted within this scope. Continued development of open benchmarks and models is essential for advancing robust and generalizable approaches for drug discovery. We invite the community to evaluate their own models under this benchmark suite and to explore MAMMAL on new datasets and tasks.

Methods

The MAMMAL method is built around three core components: The model architecture, the molecular prompt syntax, and extensive pretraining. In Subsection 4.1, we detail the architecture and its enhancements to the standard transformer.

Subsections 4.2 and 4.3 focus on the molecular representations and prompt syntax, key features that enable the support of a diverse range of pretraining and downstream tasks for drug discovery. Finally, Subsection 4.4 outlines the pretraining process, which facilitates leveraging large, cross-domain datasets and handling multiple entities simultaneously.

MAMMAL Architecture

The MAMMAL architecture builds upon the transformer architecture introduced by Vaswani et al.⁸⁷ and draws inspiration from the T5

framework⁸⁸ while introducing several key modifications on top of it. This design conceptualizes tasks as sequence-to-sequence problems within a unified model, introducing three key features:

- **Optimized for Representation and Generation Tasks:** Biomedical models for drug discovery must perform diverse tasks, ranging from token-level or prompt-level representation to tasks requiring strong generative capabilities. For instance⁸⁹, introduced a non-causal, decoder-only protein language model optimized for both understanding and generating protein-related data, outperforming state-of-the-art baselines. MAMMAL is designed to jointly optimize for such diverse tasks, enabling the model to learn from multiple stages of the drug discovery pipeline. This collective learning is facilitated by a shared encoder, which bridges gaps between distinct AI applications in drug discovery. Specifically, MAMMAL supports both encoder-only and encoder-decoder autoregressive modes. Encoder-only mode excels in representation-heavy tasks, such as classification and regression, while encoder-decoder mode is better suited for generative tasks. By sharing encoder stack weights across these modes, MAMMAL facilitates efficient multi-task training, with parameter updates conducted through gradient accumulation across all tasks.
- **Flexible Multi-Domain Structured Prompts:** Drug discovery tasks often require prompts that describe molecular complexes composed of multiple entities. These prompts may also include attributes related to the entire complex or specific internal entities to improve prediction accuracy. To address this, MAMMAL employs a modular tokenizer that enables multi-domain structured prompts by assigning distinct sub-tokenizers to each entity type. Compared to free-text prompts, structured prompts provide a consistent and explicit representation of input data, capturing relationships and attributes between entities, and facilitating the training. While MAMMAL primarily focuses on structured input, incorporating free text as an additional data source represents a natural extension. Recent advancements in biomedical large language models^{90–92}, demonstrate the potential of free-text prompts to enhance versatility and expand training datasets. Such integration could integrate free text directly, mix it with structured prompts, or leverage embeddings derived from extensively pre-trained text models⁹³, <https://github.com/ibm-granite/granite-3.0-language-models/>. These embeddings could harness the latent knowledge encoded within external models to enrich the system's performance.
- **Numerical Values Integration:** A notable innovation of MAMMAL is its native support for numerical values (e.g., 2, 3.14, 10,000.1) as both inputs and outputs. Numerical data is crucial for modeling drug discovery tasks and allows for direct representation of important attributes. This capability expands the range of tasks MAMMAL can handle, such as binding affinity prediction and protein folding. Many biomedical large language models either lack support for numerical data⁹⁰, employ limited solutions such as binning⁹², or rely on digit-based representations⁹¹, which can inflate input length⁹⁴, offer compelling digit-based approach of splitting the number into semantic parts (mainly digits), supporting arbitrary scalar values while adding only a limited number of tokens to the vocabulary. While this simplifies integration with standard language models, it inflates the number of input/output tokens significantly. This is especially evident in tasks like gene expression prediction which require supporting thousands of scalars in a single prompt and expected model output⁹⁵, suggests a new loss term that considers numerical proximity, with the main advantage of not requiring any additional prediction head for scalars, allowing to introduce numerical values support within existing language model architectures. However, it is currently limited to a fixed vocabulary, not allowing support for arbitrary (possibly unseen in train time) scalar values. Other methods, such as discretization, demonstrated in⁸¹ through pre-trained VQ-VAE⁹⁶ that encodes and decodes 3D spatial positions, represent another approach limited to a specific use case. MAMMAL, however, integrates continuous numerical values directly into its embedding space via a projection layer. The resulting

Table 3 | Pretraining Tasks

Name	Domain	Entity Type	Task Type	Dataset	Number of Samples
Protein LM	Biologic	General Protein	Spans Masking LM	Uniref90 ⁹⁷	180M
Antibody LM	Biologic	Antibody	Spans Masking LM	OAS ⁹⁸	650M
Small Molecule LM	Small Molecules	Small Molecule	Spans Masking LM	ZINC ¹⁰⁰ + PubChem ¹⁰¹	200M
Cell Genes LM	Single Cell Transcript- omics	Cell Genes	Spans Masking LM	CELLxGENE ¹⁰³	30M
Protein-Protein Interaction	Biologic	General Protein	Classification	STRING ¹⁰²	780M
Protein-Protein Interaction Gen.	Biologic	General protein	Generation	STRING ¹⁰²	390M
Antibody Denoise	Biologic	Antibody	Denoise Sequence	OAS ⁹⁸	650M

Details on the pretraining tasks that were used while training `ibm/biomed.omics.bl.sm.ma-ted-458m`. "Number of Samples" lists the post-filtering number of samples actually used. A single model was pretrained with all of the listed tasks, accumulating knowledge spanning multiple domains.

embeddings align with the input token embeddings, ensuring simple, efficient, and effective utilization of numerical data.

Further details about the MAMMAL architecture can be found in Supplementary Figure S1.

Entity Representation

Each entity domain supported by MAMMAL makes use of a representation approach that has been selected as follows:

- **Small Molecules.** Represented using SMILES sequences (Simplified Molecular Input Line Entry System). For example, paracetamol is represented as CC(=O)NC1=CC=C(O)C=C1.
- **Gene Expression.** Represented as an ordered list of gene names, sorted in descending order based on log-normalized and binned expression values. In case of ties, genes are sorted alphabetically. This representation applies to both single-cell and bulk RNA expression data.
- **Proteins.** Represented as concatenated amino acid chains, preserving original source-data order. This representation contains only sequence data, without structural information.
- **Antibodies.** Represented similarly to proteins, using concatenated amino acid sequences. Each chain is prefixed with a token indicating its type (heavy or light).

Prompt Syntax

The prompt syntax is built around a modular tokenizer architecture that governs how molecular data is interpreted and encoded. At its core, the tokenizer defines a common syntax using special tags that represent molecular entities, sequences, attributes, and interactions across diverse molecular systems. It supports multi-domain inputs by delegating different segments of the input sequence to specialized sub-tokenizers, each responsible for handling a specific data type or modality.

All sub-tokenizers operate under the umbrella of the main tokenizer and share access to a unified set of special tokens, such as `<EOS>`, ensuring consistency across domains. Among these sub-tokenizers, a designated numeric sub-tokenizer is used for handling continuous values. Instead of tokenizing numeric values into discrete tokens, this sub-tokenizer projects them directly into the model's embedding space via a learned projection layer.

The prompt syntax applies uniformly to both inputs and outputs of the model. It is also designed to be extensible: New tags can be introduced into individual sub-tokenizers or added to the shared token set without disrupting existing functionality. Thanks to this modular structure, updates to domain-specific sub-tokenizers or the token vocabulary maintain backward compatibility, enabling interoperability between newly trained and previously deployed models. More detailed explanations and examples of the prompt syntax can be found in Supplementary Figures S2, S3 and Supplementary Table S1.

Pretraining

MAMMAL is designed as a comprehensive foundation model, capable of spanning multiple domains and accommodating a variety of entities. It is intended to support diverse task types, ranging from representation-focused tasks to generation-oriented ones. To achieve this, MAMMAL is trained on multiple tasks concurrently. Pretraining was conducted on two billion samples sourced from six datasets, which are all publicly available, covering three distinct domains across seven tasks. Table 3 summarizes these tasks, detailing the relevant domains, entity types, and specific datasets. Additional details about the pretraining are provided in the Supplementary Information.

Data availability

All datasets used in this study are publicly available. **Cell Type.** The Zheng68k dataset was obtained from <https://www.10xgenomics.com/datasets/fresh-68-k-pbm-cs-donor-a-1-standard-1-1-0> (file: https://cf.10xgenomics.com/samples/cell-exp/1.1.0/fresh_68k_pbmc_donor_a/fresh_68k_pbmc_donor_a_filtered_gene_bc_matrices.tar.gz) **BBBP and ClinTox** benchmarks were obtained from <https://github.com/IBM/molformer/tree/main/data> that points to <https://ibm.ent.box.com/v/MoLFormer-data> (file: `finetune_datasets.zip`). **Cancer-Drug Response 1 and 2.** The GDSC1 and GDSC2 benchmarks were accessed with random splits from the TDC library (<https://pytorch.org/project/PyTDC/>). **Cancer-Drug Response 3.** The benchmark was obtained from the DeepCDR³⁶ git repository (<https://github.com/kimmo1019/DeepCDR/tree/master/data>). **DTI.** This benchmark was published by⁶⁰ and is available in <https://torchdrug.ai/docs/api/datasets.html#bindingdb>. **Ab Infilling.** This data was taken from⁴⁷, which provides a preprocessed subset of the publicly available SAbDab database⁵³. The preprocessing pipeline includes a similarity-based clustering for the data splits and sample-level filtering that excludes samples considered invalid in⁴⁷. For additional information, we refer to⁴⁷ and the publicly available codebase: <https://github.com/THUNLP-MT/dyMEAN>. **PPIΔΔG.** The SKEMPI S1131 dataset of non-redundant single mutations was derived from SKEMPI⁵⁵ in⁹⁷ and can be downloaded from <https://zhanggroup.org/BindProfX/download/>. **TCR Bind** The Weber TCR binding dataset was downloaded from https://tdcommons.ai/multi_pred_tasks/tcrepitopeAntibody-Antigen-Bind. The HER2 antibody-antigen binding dataset was taken from the original paper⁶² github repository https://github.com/dahjan/DMS_opt. **Pretraining.** The `ibm/biomed.omics.bl.sm.ma-ted-458m` was pre-trained over OAS⁹⁸, UniProt⁹⁹, Zinc¹⁰⁰, PubChem¹⁰¹, STRING¹⁰² and CELLxGENE¹⁰³. Supplementary Information describes the pre-processing steps applied.

Code availability

The model architecture, fine-tuning framework, and end-to-end examples are publicly available at: <https://github.com/BiomedSciAI/biomed-multi-alignment>. This GitHub repository provides comprehensive resources for working with the model, including setup instructions, fine-tuning

guidelines, and inference workflows for multiple downstream tasks. Pre-trained model weights and the associated tokenizer are hosted on the Hugging Face Model Hub and can be accessed at: <https://huggingface.co/ibm/biomed.omics.bl.sm.ma-ted-458m>. Additionally, selected fine-tuned model checkpoints and tokenizers are available via: https://huggingface.co/models?other=base_model:finetune:ibm-research/biomed.omics.bl.sm.ma-ted-458m, and can be explored interactively through the Hugging Face Space at: <https://huggingface.co/spaces/ibm/biomed-multi-alignment>.

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Study Design: Y. Shoshan, M.R., and M.O.F. contributed to the conception and design of the study. AI Development: Y. Shoshan, M.R., M.O.F., V.R., A.G., E.B., S.R.C., S.P., I.A., B.S., L.H., M.N., S.R., and M.D. were involved in the development of the AI components used in the study. Experiments: M.R., V.R., A.G., E.B., I.A., Y. Shamay, and S.K. conducted the experiments and contributed to data acquisition and analysis. Manuscript Writing: Y. Shoshan., M.R., M.O.F., V.R., A.G., E.B., S.R.C., S.K. and Y. Shamay contributed to drafting and revising the manuscript. Critical Revisions: J.W., J.M., and P.S. provided substantial edits and critical feedback on the manuscript. Supervision: E.H. and M.R.Z. supervised the overall project and provided strategic guidance. All authors have read and approved the submitted version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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